DEPARTMENT OF ENVIRONEMNTAL ENGINEERING

REMOTE SENSING AND GEOGRIPHICAL INFORMATION SYSTEM

MICROPROJECT

MAPPING OF PM2.5 AND PM10 POLLUTANTS IN JHARKHAND

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(23/EN/107)

MICROPROJECT

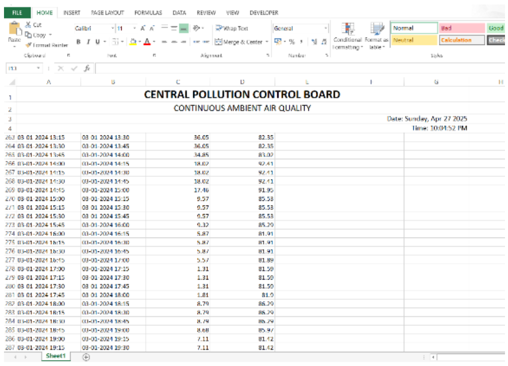
MAPPING OF PM2.5 AND PM10 POLLUTANTS IN JHARKHAND

Objective: The aim of this project is to visualize and analyze the distribution of PM2.5 and PM10 pollutants across various cities in Jharkhand using QGIS software. This project will help in understanding the spatial pattern of air pollution and identifying the areas most affected.

Introduction: Air pollution is a critical issue in India, and particulate matter (PM2.5 and PM10) are among the most harmful pollutants. Mapping these pollutants spatially allows researchers and policymakers to assess the air quality and plan mitigation strategies effectively. Jharkhand, being a mineral-rich state, faces significant pollution due to mining and industrial activities.

Materials Required:

- QGIS Software (Version 3.22 or later)
- CSV file containing data: City name, Monitoring Station, Latitude, Longitude, PM2.5, and PM10 concentrations
- Jharkhand district shapefile (boundary layer)



CENTRAL POLLUTION CONTROL BOARD

CONTINUOUS AMBIENT AIR QUALITY

Date: Sunday, Apr 27 2025

Time: 10:04:52 PM

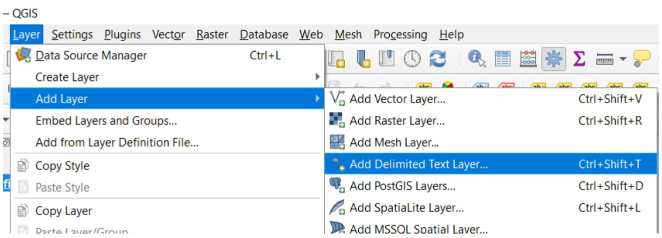
	A	B	C	D
263	03-01-2024 13:15	03-01-2024 13:30	36.05	82.35
264	03-01-2024 13:30	03-01-2024 13:45	36.05	82.35
265	03-01-2024 13:45	03-01-2024 14:00	34.85	83.02
266	03-01-2024 14:00	03-01-2024 14:15	18.02	92.41
267	03-01-2024 14:15	03-01-2024 14:30	18.02	92.41
268	03-01-2024 14:30	03-01-2024 14:45	18.02	92.41
269	03-01-2024 14:45	03-01-2024 15:00	17.46	91.95
270	03-01-2024 15:00	03-01-2024 15:15	9.57	85.53
271	03-01-2024 15:15	03-01-2024 15:30	9.57	85.53
272	03-01-2024 15:30	03-01-2024 15:45	9.57	85.53
273	03-01-2024 15:45	03-01-2024 16:00	9.32	85.29
274	03-01-2024 16:00	03-01-2024 16:15	5.87	81.91
275	03-01-2024 16:15	03-01-2024 16:30	5.87	81.91
276	03-01-2024 16:30	03-01-2024 16:45	5.87	81.91
277	03-01-2024 16:45	03-01-2024 17:00	5.57	81.89
278	03-01-2024 17:00	03-01-2024 17:15	1.31	81.59
279	03-01-2024 17:15	03-01-2024 17:30	1.31	81.59
280	03-01-2024 17:30	03-01-2024 17:45	1.31	81.59
281	03-01-2024 17:45	03-01-2024 18:00	1.81	81.9
282	03-01-2024 18:00	03-01-2024 18:15	8.79	86.29
283	03-01-2024 18:15	03-01-2024 18:30	8.79	86.29
284	03-01-2024 18:30	03-01-2024 18:45	8.79	86.29
285	03-01-2024 18:45	03-01-2024 19:00	8.68	85.97
286	03-01-2024 19:00	03-01-2024 19:15	7.11	81.42
287	03-01-2024 19:15	03-01-2024 19:30	7.11	81.42

Sheet1

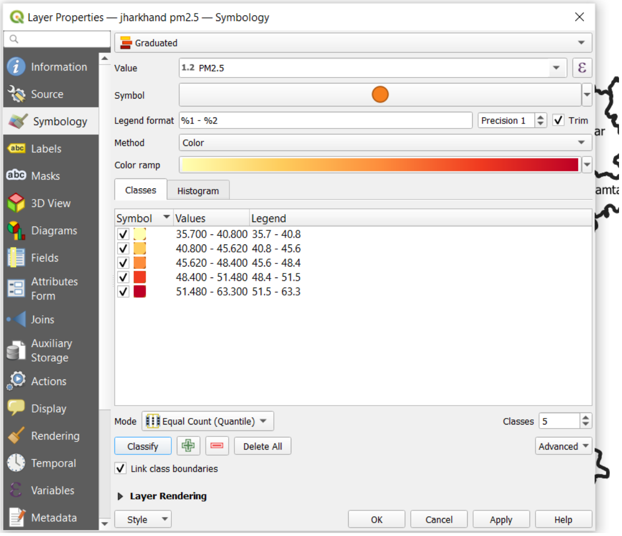
<u>Methodology:</u>

1. **Data Collection:**
   - Gather PM2.5 and PM10 data for major cities in Jharkhand.
   - Collect the geographic coordinates (latitude and longitude) of monitoring stations.

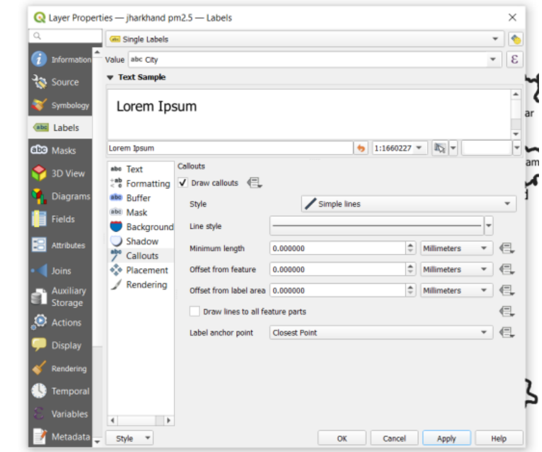
2. **Setting Up QGIS Project:**
   - Load the Jharkhand shapefile into QGIS.
   - Import the CSV file as a point layer (using 'Add Layer' -> 'Add Delimited Text Layer').



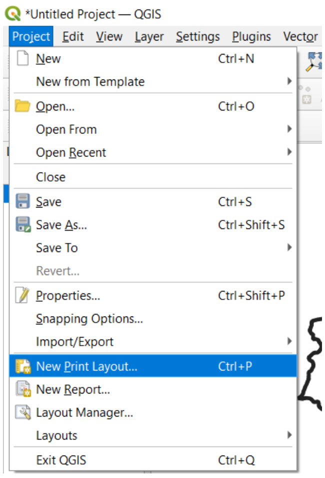
3. **Symbolizing the Pollutant Data:**
   - For PM2.5:
     - Right-click the point layer -> Properties -> Symbology.
     - Set 'Graduated' style based on PM2.5 values.
     - Apply a color ramp from Yellow (low) to Red (high).
   - For PM10:
     - Repeat the process by creating a duplicate layer styled for PM10 values.

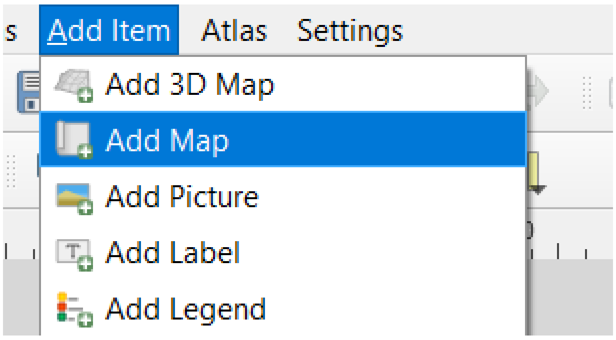


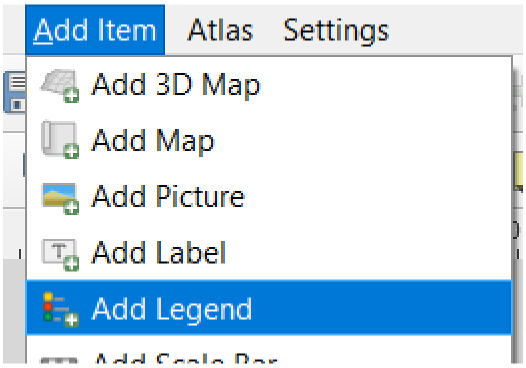
4. **Adding Labels:**
   - Label points with the name of the monitoring stations/cities
   - Adjust label settings (font size, buffer, placement) for better readability.



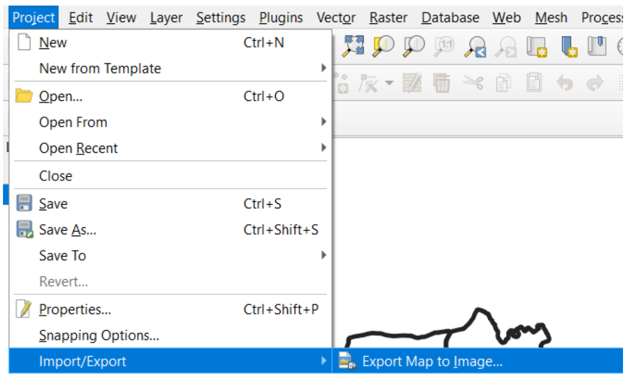
5. **Creating the Layout:**
   - Open a New Print Layout.
   - Add the map, legend, north arrow, and scale bar.
   - Customize the legend to reflect PM2.5 and PM10 concentrations clearly.







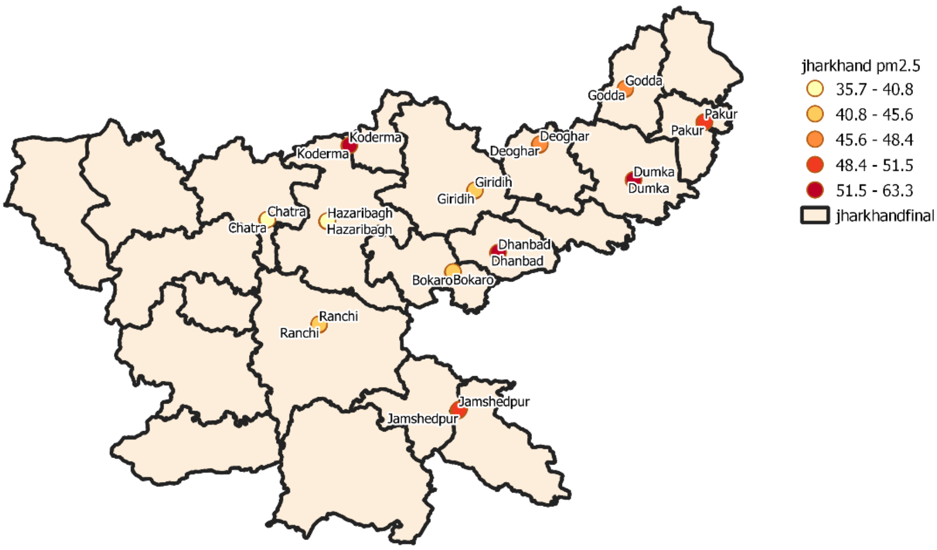
6. **Final Map Export:**
    - Export the final map as PDF or image for presentation.



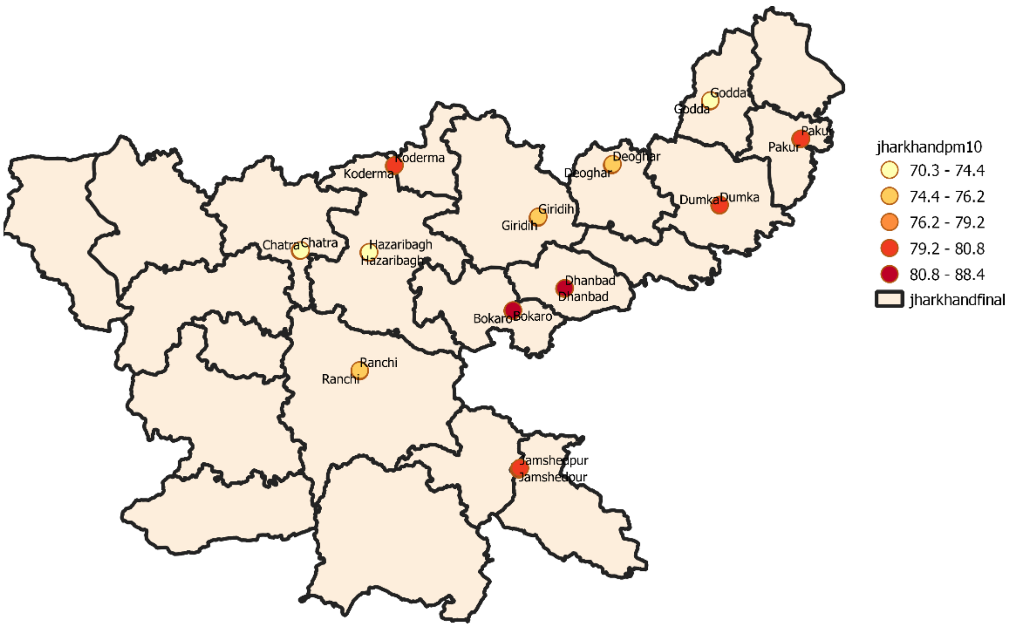
Observations:

- Areas such as Dhanbad and Dumka show higher PM2.5 and PM10 levels, likely due to industrial emissions and mining activities.
- Cleaner areas are observed in districts with more forest cover and less industrial activity.

PM-2.5



PM-10



CONCLUSION

The mapping and spatial analysis of PM2.5 and PM10 pollutant concentrations across cities of Jharkhand using QGIS provided valuable insights into the region's air quality status. Through this study, we identified key pollution hotspots, highlighted regional variations, and recognized areas with comparatively cleaner air.

The project demonstrated the power of Geographic Information Systems (GIS) in visualizing environmental data, making complex patterns easily understandable. It also emphasized the urgent need for sustainable urban planning, stricter pollution control measures, and regular monitoring to ensure better air quality across Jharkhand.

Future studies could include temporal analysis (monthly/seasonal variation), incorporation of meteorological parameters, and real-time monitoring for more dynamic and responsive environmental management. This microproject not only supports environmental awareness but also strengthens the use of technology-driven solutions for public health and urban development challenges.

Future Scope:

1) Time-Series Analysis

- Extend the study by analyzing seasonal and yearly trends in PM2.5 and PM10 levels to understand how air quality changes over time.

2) Integration with Meteorological Data

- Combine pollutant data with weather parameters (wind speed, humidity, temperature) to better predict pollution dispersion patterns.

3) **Real-Time Air Quality Mapping**

- Incorporate live sensor data and IoT technologies to develop dynamic, real-time air quality maps for public information and decision-making.

REFERENCES

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5) www.onlinemaps.surveyofindia.org

6) www.stategisportal.nic.in

7) www.gis.jharkhand.gov.in

8) www.divagis.org.in

9) www.gadm.org

10) www.simplemaps.com